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Review

From Sustainability to Regeneration: A Systems Approach to Mountain Forestscape Restoration

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Abstract: Mountain forestscapes are among the planet's most vital socio-ecological systems, functioning as critical reservoirs of biodiversity, regulators of climate, and essential sources of water and livelihood for surrounding and downstream communities. However, these landscapes face intensifying threats from climate change, land-use transformation, resource extraction, unsustainable tourism, and fragmented governance. While sustainable forest management has provided essential frameworks for conservation, its focus on maintaining existing conditions is increasingly insufficient amid accelerating ecological and social decline. This paper explores the conceptual and practical evolution from sustainability to regeneration in mountain forest management, distinguishing between the theoretical foundations of each paradigm and analyzing the key drivers of degradation across diverse biocultural regions. Methodologically, the study employs a transdisciplinary and qualitative design, integrating critical literature synthesis, comparative regional analysis, and Indigenous and local knowledge systems. A structured review of 72 peer-reviewed sources and ethnographic materials was conducted, combined with a qualitative comparative analysis (QCA) of mountain case studies from the Alps, Carpathians, Andes, Eastern Arc Mountains, and Himalayas. The study synthesizes regenerative strategies such as ecological connectivity restoration, agroecological landscape design, participatory watershed management, and adaptive, polycentric governance. Based on these insights, an integrated framework for regenerative mountain forestscape management is proposed, aligned with global policy agendas including the UN Decade on Ecosystem Restoration, nature-based solutions (NbS), and the post-2020 Global Biodiversity Framework. By prioritizing ecological renewal, cultural continuity, and community agency, this work contributes to transformative, place-based approaches that restore the functionality, resilience, and integrity of mountain landscapes.

Keywords: ecosystem restoration; forest management; mountain ecosystems; regenerative ecology; socio-ecological resilience; sustainability

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Academic Editors: Fausto Sarmiento, Barbaros Gönençgil and Pankaj Kumar

Received: 3 April 2025
Revised: 22 April 2025
Accepted: 26 April 2025
Published: 29 April 2025

Citation: Majlingova, A. From Sustainability to Regeneration: A Systems Approach to Mountain Forestscape Restoration. *Sustainability* **2025**, *17*, 4001. https://doi.org/10.3390/su17094001

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1. Introduction

1.1. Background

Mountain forestscapes are among the most ecologically critical and environmentally sensitive regions on Earth. Distributed across diverse latitudinal and elevational gradients, they act as biodiversity hotspots, carbon sinks, and key freshwater sources—earning the title of the planet's "water towers" [1,2].

These ecosystems exhibit high levels of species richness and endemism, especially in montane regions such as the Andes, Himalayas, Eastern Arc, and Albertine Rift [3,4]. Their complex vertical zonation supports a mosaic of microclimates and ecological niches, making

them integral to global biodiversity patterns and ecosystem functioning [5]. Mountain forests also contribute significantly to climate regulation by sequestering carbon, regulating hydrological cycles, and influencing atmospheric moisture and temperature dynamics [6,7].

They serve as natural buffers against climate extremes by stabilizing slopes, reducing erosion, and maintaining the flow of water downstream. As headwaters for major river systems such as the Amazon, Yangtze, Nile, and Ganges, mountain forestscapes are crucial to agriculture, hydropower, and drinking water supply across vast geographic areas [8,9].

Furthermore, they underpin the cultural and economic well-being of Indigenous and local communities who manage these landscapes through long-standing agroecological and spiritual traditions. These communities often act as stewards of regeneration, yet remain vulnerable to land-use change, ecological degradation, and climate-induced risks [10,11].

Preserving and regenerating mountain forestscapes is therefore a strategic priority for biodiversity conservation, climate mitigation, and sustainable development globally [12,13].

Pressures: Climate Change, Tourism, Logging, Shifting Cultivation, etc.

Mountain forestscapes are increasingly exposed to a convergence of pressures that undermine their ecological integrity, socio-cultural resilience, and long-term sustainability. These pressures—both external and internal—interact across scales and intensify existing vulnerabilities. The primary drivers include climate change, deforestation from logging, unsustainable tourism, agricultural expansion including shifting cultivation, and weak governance frameworks.

Climate change is perhaps the most pervasive pressure, altering temperature regimes, precipitation patterns, and seasonal cycles across mountain ecosystems. Evidence shows significant upslope migration of species, glacier retreat, and increasing frequency of extreme events such as landslides, droughts, and wildfires [14,15].

Mountain forests are particularly sensitive to warming trends. For instance, the Himalayas have warmed nearly twice as fast as the global average, threatening treeline dynamics, water supply, and ecosystem stability [16]. Climate-induced phenological shifts also disrupt plant-pollinator interactions, carbon cycling, and biodiversity distributions [17].

Commercial and illegal logging continue to be major drivers of forest degradation in mountainous regions. Often facilitated by weak enforcement and lack of tenure clarity, logging activities lead to fragmentation, loss of habitat connectivity, and biodiversity decline [18]. Road construction for timber extraction also opens remote areas to further exploitation. These activities are often driven by short-term economic goals and poorly regulated land governance.

In places like the Eastern Himalayas and Tropical Andes, the expansion of logging roads correlates with increased deforestation and forest edge effects, impacting endemic species and carbon storage capacity [19,20].

Tourism in mountain areas has grown rapidly, often outpacing environmental safeguards. While ecotourism presents opportunities for conservation-based livelihoods, mass tourism contributes to waste accumulation, land conversion, water stress, and cultural commodification [21].

High-altitude destinations like the Swiss Alps, Machu Picchu, and the Mount Everest region have witnessed ecosystem stress from trail erosion, deforestation for infrastructure, and increased human–wildlife conflict [22]. Tourism-related infrastructure (e.g., roads, resorts, ski slopes) accelerates habitat fragmentation and disrupts ecological corridors [23].

Agricultural encroachment, particularly shifting cultivation (slash-and-burn), remains a key driver of degradation in many tropical and subtropical mountains. While shifting cultivation has historically been sustainable in low-density contexts, increased demographic

pressure and shortened fallow cycles have made the practice ecologically destructive in some areas [24].

Mountain slopes converted for monoculture crops or grazing often result in soil erosion, nutrient depletion, and reduced forest regeneration. In the Eastern Arc Mountains of Tanzania and the uplands of Southeast Asia, land-use change for agriculture has led to significant forest loss and biodiversity decline [25,26].

Underlying many of the above pressures is the lack of effective governance, secure land tenure, and inclusive decision making. Insecure land rights discourage long-term stewardship and create conflicts between local communities and external actors, including corporations and state agencies [27].

Moreover, policy fragmentation and a top-down conservation model have historically marginalized Indigenous knowledge and regenerative practices [28]. Without integrated governance frameworks that blend scientific, local, and Indigenous perspectives, forest degradation is likely to continue despite restoration efforts.

Although degradation drivers in mountain forestscapes share commonalities, the intensity and combinations of pressures vary significantly by region, shaped by biogeography, governance regimes, economic structures, and cultural dynamics.

In the Himalayas, climate change is intensifying glacier retreat, altering precipitation patterns, and threatening slope ecosystems. Simultaneously, unregulated tourism and road expansion in ecologically fragile zones contribute to deforestation, erosion, and wildlife disturbance [29,30]. Illegal logging, often tied to poorly enforced forest governance, further exacerbates fragmentation, particularly in the Eastern Himalayan region [31].

In the Tropical Andes, deforestation for agriculture—particularly coffee, cocoa, and pastureland—has been a dominant driver, leading to the degradation of cloud forests and montane habitats. Illicit crop cultivation, mining, and road building further compound forest loss in Colombia, Ecuador, and Peru [32,33]. Climate change adds pressure by disrupting hydrological flows and reducing páramo resilience [34].

The Eastern Arc Mountains of East Africa, a recognized biodiversity hotspot, face pressures from shifting cultivation, firewood collection, and charcoal production. These drivers are compounded by population density and poverty, leading to shortened fallow periods, reduced regeneration capacity, and forest fragmentation [35,36].

In Southeast Asia's upland areas—notably northern Laos, Vietnam, and Myanmar—shifting cultivation has transitioned from traditional rotational systems to more permanent forms of land use, driven by demographic pressures and commercial agriculture. This transition, coupled with infrastructure development, has contributed to deforestation, erosion, and biodiversity loss [37,38].

In contrast, European mountain forestscapes, particularly in the Alps, Carpathians, and Pyrenees, face a different set of pressures linked to land abandonment, land-use intensification, and climate change. Following widespread rural depopulation in the 20th century, many subalpine agricultural areas experienced natural forest regrowth [39]. While this secondary succession has enhanced carbon storage and connectivity in some areas, it has also led to the homogenization of landscapes, a decline in traditional agro-pastoral systems, and the loss of open-habitat species [40].

Climate change is shifting treelines upward and altering snowpack dynamics in the Alps, affecting forest structure, species distribution, and ecosystem services [41]. Meanwhile, tourism infrastructure—especially ski resorts, cableways, and associated roads—has resulted in fragmentation of alpine habitats, soil compaction, and vegetation disturbance [42].

The Carpathians, which contain Europe's largest remaining tracts of old-growth forest, face pressures from illegal logging, poorly monitored timber harvesting, and weak trans-

boundary governance [43]. Fragmentation from road networks and unregulated tourism is also increasing, despite international conservation efforts like the Carpathian Convention.

Together, these cases illustrate how degradation drivers in mountain forestscapes are highly contextual. Socioeconomic transitions, climatic variability, and governance capacities shape the pathways and pace of degradation, necessitating adaptive, region-specific responses that account for both ecological and cultural diversity.

1.2. Conceptual Clarification of Key Terms

To enhance conceptual clarity, Table 1 provides definitions for frequently used terms within this manuscript:

Table 1. Definitions for frequently used terms.

Term	Definition
Regeneration	A holistic process aimed at restoring the ecological function, cultural integrity, and social relationships within ecosystems. Goes beyond sustainability by enabling co-evolution and renewal.
Sustainability	Maintenance of resource use and ecosystem conditions to meet present and future needs without degradation. Often focuses on minimizing harm.
Revitalization	Cultural and ecological renewal, often through the restoration of traditional practices and community-led initiatives.
Resilience	The capacity of socio-ecological systems to absorb disturbances, reorganize, and retain essential structure and functions.

Distinction Between "Sustainable" and "Regenerative" Approaches

In the context of complex and dynamic systems such as mountain forestscapes, the conceptual divergence between sustainability and regeneration reflects a critical shift in environmental thinking. While sustainability has long provided a foundational framework for environmental stewardship, the growing ecological crisis has prompted a move toward more transformative, regenerative paradigms that emphasize system renewal and socio-ecological co-evolution [44–46].

Sustainability is rooted in the principle of maintaining ecological balance and ensuring the long-term availability of resources for present and future generations. It tends to focus on conservation, efficiency, and resilience within existing system boundaries [47]. In forest landscapes, this translates into practices such as controlled logging, forest certification, reforestation, and ecosystem service valuation [48].

However, in many cases—especially in mountain regions where systems are already degraded or shifting rapidly due to climate change—sustainability may not suffice. Efforts to maintain a degraded baseline or to "do less harm" often fall short of addressing underlying drivers of ecological collapse or social inequity [49].

Regeneration, by contrast, seeks not only to preserve but also to revitalize and enhance ecosystems. It recognizes that degraded environments require more than maintenance—they demand active healing, renewal, and co-creation with nature [15]. Regenerative approaches aim to increase biodiversity, rebuild ecosystem functions, restore nutrient cycles, and empower communities as agents of ecological transformation [10,14].

This orientation aligns with systems thinking and complexity science, viewing landscapes as dynamic, living systems that can evolve toward higher levels of health and functionality when supported by appropriate human relationships [50]. Rather than focusing solely on minimizing impact, regeneration asks how human activity can create net-positive outcomes for ecosystems and communities alike [48].

The sustainability-regeneration distinction becomes particularly significant in mountain forestscapes. These ecosystems are undergoing disproportionate impacts from climate

change—including tree line shifts, glacier loss, and species migration—coupled with increasing human pressures from tourism, agriculture, and resource extraction [39,43].

Sustainable forest management in these regions might involve slope stabilization, fire prevention, and water conservation. Regenerative management, however, would go further by restoring ecological connectivity, revitalizing traditional agroecological mosaics, and integrating Indigenous and local knowledge systems into adaptive co-governance models [41,51].

In this sense, regeneration is not an alternative to sustainability but a deepening and evolution of it. Where sustainability seeks to "sustain what is", regeneration seeks to transform what can be, especially in landscapes already in decline or transition.

1.3. Aims of the Study

This paper responds to the critical shift by aiming to:

- Clarify the conceptual distinction between sustainable and regenerative approaches to forest management, with specific relevance to mountain socio-ecological systems;
- assess the current pressures and systemic drivers of degradation in mountain forestscapes, including climate change, land-use transformation, biodiversity loss, and institutional fragmentation;
- synthesize regenerative principles and practices informed by ecological science, Indigenous knowledge, and biocultural frameworks that enable renewal of both ecosystem functions and cultural–ecological relationships;
- explore governance innovations and implementation models that support regenerative forest management, such as polycentric governance, participatory planning, and adaptive co-management;
- propose an integrated, future-oriented framework for regenerative mountain forestscape management aligned with global agendas like the UN Decade on Ecosystem Restoration, nature-based solutions (NbS), and the post-2020 Global Biodiversity Framework.

Through these objectives, the paper seeks to inform the development of transformative, place-based strategies that foster ecological resilience, social equity, and long-term sustainability in mountain regions experiencing rapid change.

2. Materials and Methods

This study employs a transdisciplinary and qualitative research design, appropriate for conceptual inquiry into emerging ecological paradigms and their application in complex socio-ecological systems such as mountain forestscapes. The methodological framework integrates critical literature synthesis, comparative regional analysis, and knowledge system integration, with the aim of addressing both theoretical and practice-oriented objectives.

To explore the evolving discourse from sustainability to regeneration, the study draws on an interpretive, theory-building approach based on critical review methodology [52]. This involved identifying, selecting, and synthesizing peer-reviewed publications, policy documents, and grey literature that contribute to the development of ecological thought, particularly regarding:

- Sustainable forest management (SFM) and its limitations.
- Regenerative design and ecological thinking.
- Systems theory, socio-ecological resilience, and adaptive governance.
- Biocultural diversity and Indigenous ecological knowledge.

The theoretical synthesis was iterative and abductive, moving between emerging frameworks and real-world applications to build a conceptual scaffold for regenerative forest management in mountainous regions.

To operationalize the qualitative comparative analysis (QCA), we constructed a case matrix that captured four dimensions: degradation drivers, governance type, regeneration strategies, and observed outcomes. Each case was coded for outcome indicators (e.g., canopy recovery, species richness, governance inclusion) using a thematic analysis approach. The QCA enabled cross-regional synthesis of patterns and divergences. Indigenous and local knowledge (ILK) sources were selected based on criteria including peer-review status, community authorship or co-authorship, and citation in regional or global conservation reports. A supplementary Appendix A provides a table summarizing the selection process for ILK and regional cases. A structured search strategy was conducted using the Web of Science Core Collection, Scopus, and ScienceDirect, targeting literature published between 2010 and 2024.

The selection process followed defined inclusion and exclusion criteria, summarized in Table 2. For Table 2, steps 2, 3, and 4 involved qualitative screening based on keyword relevance in titles/abstracts, regional tagging of content (e.g., Andes, Himalayas), and peer-review validation. Literature relevance to regeneration was assessed through text mining of terms (e.g., "restoration", "connectivity", "biocultural", "adaptive governance"). A list of the 326 initial and 72 final sources was used for this purpose.

Table 2. Defined inclusion and exclusion criteria for selection process.

Step	Criteria Applied	Outcome
1	Keyword search (e.g., "regenerative forestry", "mountain ecosystems", "Indigenous forest governance")	326 initial results
2	Filtered for relevance to mountain forestscapes and regeneration	210 articles selected for review
3	Screened for peer-reviewed status and regional balance	138 studies retained
4	Focused selection of theoretical and case-based studies	72 sources synthesized

This matrix ensured methodological transparency and thematic consistency across the synthesis.

Inclusion criteria focused on:

- Relevance to mountain ecosystems and forest landscapes.
- Engagement with regenerative or sustainability transitions.
- Emphasis on socio-ecological systems and participatory governance.
- Presence of regional or thematic case studies.

Keywords included combinations of: mountain forests, regeneration, sustainability, adaptive governance, biocultural diversity, and Indigenous knowledge. Sources were screened for conceptual clarity, methodological transparency, and regional relevance.

To complement the conceptual analysis, the study undertook a qualitative synthesis of published case studies from representative mountain regions. Rather than a meta-analysis, this approach follows qualitative comparative analysis (QCA) principles to explore how degradation drivers and regenerative interventions manifest under different ecological, political, and cultural conditions.

Selected regions include (but are not limited to):

- The Alps (Central Europe)
- The Carpathians (Eastern Europe)
- The Himalayas (South Asia)
- The Andes (South America)
- The Eastern Arc Mountains (East Africa)

Cases were analyzed according to a consistent matrix capturing:

- Type and intensity of degradation drivers.

- Role of local and Indigenous governance.
- Regenerative strategies applied or proposed.
- Observed socio-ecological outcomes.

This comparative approach enables the identification of patterns and divergences across biocultural landscapes, informing the paper's integrative framework.

Recognizing the epistemic contributions of Indigenous and local knowledge systems, the study includes a desk-based review of ILK literature sourced from ethnographic studies, community-authored reports, participatory mapping outputs, and Indigenous co-authored publications. Emphasis was placed on:

- Continuity of land-based ecological knowledge.
- Customary forest governance systems.
- Ritual, spiritual, and symbolic relationships to forestscapes.
- Knowledge-practice-belief systems supporting regeneration.

To ensure respectful and accurate representation, priority was given to sources that are peer-reviewed, community endorsed, or co-produced with Indigenous authors and organizations.

All findings were analyzed through a regenerative systems lens, integrating concepts from:

- Socio-ecological resilience theory [53].
- Biocultural diversity and relational worldviews [54].
- Adaptive governance and polycentric management [55].
- Regenerative development and ecological design [56].

The resulting framework provides a multi-level synthesis linking ecological restoration, cultural revitalization, and institutional transformation as co-dependent components of regenerative mountain forest management.

3. Results

The synthesis of scholarly literature, regional case studies, and Indigenous knowledge frameworks demonstrates an increasingly evident shift in mountain forest management from conventional sustainability approaches toward more transformative, regenerative paradigms. This section details the conceptual findings, thematic patterns, and applied outcomes derived from the study's integrative methodology.

3.1. Limitations of Sustainability in Mountain Forestscapes

The literature review reveals that conventional Sustainable Forest Management (SFM) has played a significant role in promoting forest conservation, controlling timber extraction, and mainstreaming ecosystem service valuation [48]. However, in mountain forestscapes—characterized by ecological fragility, vertical ecological gradients, and deep cultural embeddedness—several critical limitations of the sustainability paradigm are evident:

- Static Baselines and Limited Scope: Sustainability often assumes that ecological systems function optimally if maintained within historical or static baselines. This perspective struggles to address dynamic degradation processes such as landslides, glacial retreat, or novel species invasions driven by climate change [43,57].
- Technocratic Orientation: SFM typically relies on quantitative metrics (e.g., allowable cut, carbon sequestration rates), emphasizing resource optimization and efficiency over systemic renewal or relational values [39,41].
- Cultural Disembedding: Many SFM approaches are detached from the socio-cultural realities of Indigenous and local communities. They tend to marginalize customary

practices, sacred landscapes, and cultural institutions that contribute to ecological stewardship [39,56,58,59].

- Governance Centralization: Sustainability-based forestry policies often rely on hierarchical, centralized governance systems. These models lack flexibility and fail to recognize polycentric and adaptive governance mechanisms better suited to the complexity of mountain regions [14].

The cumulative effect is that while sustainability approaches may slow degradation, they rarely achieve ecological renewal, social empowerment, or cultural continuity—especially where legacies of colonialism, market-driven development, and ecological fragmentation converge.

3.2. Emergence of Regenerative Practices in Mountain Contexts

In contrast to the limitations of sustainability, a growing body of literature and field-based practice points to the emergence of regenerative approaches that center ecosystem healing, cultural revitalization, and adaptive governance.

European Context:

- In the Italian and French Alps, rewilding projects led by Rewilding Europe and local NGOs have facilitated the natural return of large herbivores, forest regeneration on abandoned pastures, and habitat connectivity for apex predators. These initiatives emphasize ecological spontaneity, low-intervention approaches, and community co-ownership [59].
- In the Carpathians, civil society organizations and community groups have begun to integrate traditional land-use patterns—such as wood pasture systems—with biodiversity corridors and riparian restoration, enhancing both landscape diversity and carbon storage [56].

Global South Context:

- In the Eastern Arc Mountains of Tanzania, participatory forest management and assisted natural regeneration (ANR) have been adopted to rehabilitate degraded forests. Community forest associations actively monitor regrowth, enforce traditional bylaws, and co-manage with government forestry officers [60].
- In the Andes, highland communities engage in mosaic agroforestry systems that integrate potato terraces, native species buffer zones, and ritual forest enclosures (e.g., "bosques de agua"). These systems support biodiversity, food sovereignty, and the cultural transmission of ecological knowledge [10].
- In the Khasi Hills (India), REDD+ pilot projects combined with clan-based forest governance have led to successful forest restoration, carbon revenue sharing, and the revitalization of sacred groves as ecological sanctuaries [11].

These examples illustrate that regenerative approaches are not uniform; they are shaped by local ecologies, cosmologies, and socio-political histories. However, they share common traits: emphasis on self-renewing systems, respect for traditional stewardship, and a long-term vision of socio-ecological health.

3.3. Core Features of Regenerative Systems: Relational, Adaptive, Biocultural

The conceptual synthesis identifies three recurring features of regenerative systems across regions, knowledge systems, and management models:

- Relationality: Regeneration reconfigures the human–nature relationship from a utilitarian model to one based on reciprocity, kinship, and care. Forests are viewed as active relatives, teachers, or ancestral domains, rather than passive resources [55,61]. This

worldview is foundational in Indigenous cosmologies but is increasingly acknowledged in regenerative design literature as essential for fostering ecological ethics.

- Adaptivity: Given the inherent uncertainty of mountain environments, adaptive capacity is critical. Regenerative systems rely on iterative learning, decentralized governance, and cross-scalar coordination (local-national-transboundary). These elements enable forest users to respond to changing climatic conditions, socio-economic shifts, and ecological surprises [14].
- Biocultural Integration: Regenerative practices consistently demonstrate that cultural heritage, land-based identity, and ecological knowledge are not peripheral but central to ecological renewal. This includes customary fire regimes, sacred forest rituals, seasonal harvesting calendars, and Indigenous toponymies that encode ecological management strategies [11,39,58,62].

3.4. An Integrated Framework for Regenerative Mountain Forestscapes

The synthesis culminates in a conceptual framework (Table 3) that integrates four interdependent dimensions of regeneration. This framework is intended to guide policy, practice, and research in transitioning from static sustainability models toward living, evolving forest systems.

Table 3. Integrated framework for regenerative mountain forestscapes.

Regenerative Dimension	Description
Ecological restoration	Rebuilding degraded ecosystems by enhancing ecological function and structure.
Cultural revitalization	Strengthening place-based identities, practices, and values tied to forest life.
Governance transformation	Fostering inclusive, adaptive, and polycentric governance arrangements.
Systemic resilience	Enhancing the capacity to absorb disturbance, reorganize, and adapt over time.

These dimensions are not sequential but interact continuously. The circular nature of this framework reflects the co-evolutionary dynamics of mountain forestscapes, where adaptive feedback loops shape socio-ecological outcomes. This integrative perspective aligns with the nature-based solutions (NbS) typologies put forth by the IUCN, which emphasize actions that deliver co-benefits for biodiversity and human well-being through ecosystem restoration and stewardship [63].

Furthermore, the framework is consistent with the IPBES Nature Futures Framework, which promotes pluralistic pathways toward biodiversity-positive futures by incorporating multiple knowledge systems, long-term visioning, and relational values [64]. Our model embeds these principles by linking ecological function with cultural memory and institutional adaptability, thereby supporting transformations that are both restorative and forward-looking.

A visual representation of this framework (Figure 1) illustrates the layered interactions between the ecological, cultural, and governance domains. The diagram presents these as overlapping circles or a tri-layered system nested within systemic resilience, symbolizing interdependence and feedback.

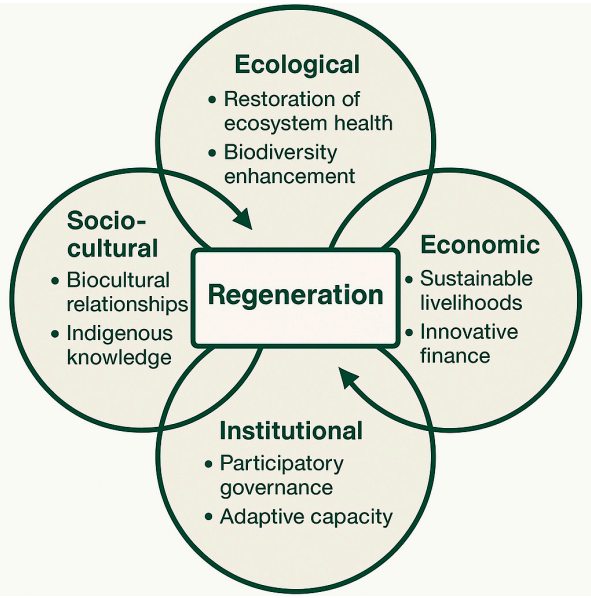


Figure 1. Conceptual framework for regenerative mountain forestscapes: interconnected pillars of ecological restoration, cultural revitalization, and governance transformation, nested within systemic resilience.

3.5. Strategic Approaches to Regenerative Forest Management

Building on the conceptual findings, this section outlines a set of actionable strategies aligned with the regenerative paradigm. These strategies integrate ecological science, traditional knowledge, and adaptive governance principles to address the multifaceted challenges of mountain forestscapes.

3.5.1. Ecological Restoration and Agroforestry

Regenerative ecological practices emphasize the restoration of forest structure, function, and biodiversity through nature-based and community-driven approaches:

- Assisted natural regeneration (ANR) has proven effective in restoring degraded forest patches, particularly in tropical mountain systems where seed banks and rootstocks remain viable. In the Eastern Arc Mountains, ANR is supported by participatory monitoring and local knowledge of forest succession [62,65].
- Native species reintroduction enhances biodiversity and resilience by restoring locally adapted plant communities. This is particularly relevant in the Alps and Carpathians, where native beech, oak, and fir species are being replanted to replace non-native monocultures.
- Agroforestry and Silvopastoral systems create multifunctional landscapes. Forest-agriculture mosaics in the Andes and Himalayas blend traditional crops, fodder trees, and native vegetation to support both livelihoods and ecosystem services [66,67].

3.5.2. Community-Based Forest Management

Inclusive governance is a cornerstone of regenerative approaches. When local communities are empowered as co-managers, forest outcomes tend to improve:

- Co-governance and Indigenous stewardship models (e.g., India's Van Panchayats, Nepal's CFUGs) have demonstrated the effectiveness of community institutions in forest regeneration, fire management, and equitable resource distribution.
- Payment for ecosystem services (PES) schemes are increasingly integrated into regenerative models. In the Khasi Hills, PES supports the conservation of sacred groves while recognizing customary tenure and enabling benefit-sharing [68].

3.5.3. Climate-Resilient Landscape Design

Designing mountain landscapes for climate resilience involves combining ecological principles with socio-cultural priorities:

- Watershed restoration improves water retention and sediment stability, reducing landslide risks and ensuring downstream water supply. Examples include reforestation along riparian zones in the Alps and Andes.
- Connectivity corridors and biodiversity refugia are essential for enabling species migration under climate stress. Rewilding Europe's initiatives in the Alpine region include reconnecting forest patches via wildlife corridors [69].
- Integrated resilience metrics are being developed to assess the co-benefits of regenerative action, combining ecological indicators (e.g., canopy cover, species richness) with social indicators (e.g., local governance quality, food security).

3.5.4. Policy and Governance Innovations

Enabling regeneration at scale requires institutional change and multi-level collaboration:

- Polycentric governance structures, where multiple authorities collaborate across sectors and scales, enhance flexibility and local adaptation. This is especially relevant in transboundary mountain regions such as the Alps and the Hindu Kush Himalaya [62].
- Landscape-level planning integrates ecological connectivity, watershed dynamics, and land-use zoning. Examples include regional landscape corridors in the Carpathians and the EU's Green Infrastructure Strategy.
- Decentralized and adaptive institutions support continuous learning and responsiveness to environmental change. Forest user groups, municipal watershed committees, and Indigenous councils offer scalable models of decentralized governance with embedded ecological knowledge.

These strategies reflect a paradigm shift—from minimizing harm to actively co-creating living, resilient landscapes that support biodiversity, cultural continuity, and climate adaptation.

3.6. Comparative Case Summary

To ground the theoretical insights in empirical detail, Table 4 summarizes key regenerative case studies by region, type of intervention, governance model, and observable outcomes.

Table 4. Key regenerative case studies.

Region	Intervention	Governance	Outcomes
Andes	Mosaic agroforestry	Community-led	Biodiversity increase, food security
Alps	Rewilding corridors	NGO–local partnerships	Habitat connectivity, predator return
Eastern Arc	ANR and bylaws	State-community co-management	Forest regeneration, reduced fire risk
Khasi Hills	PES and sacred groves	Clan-based	Cultural revival, carbon credits

3.7. Operationalizing the Framework

To guide real-world application, the framework (Table 2; Figure 1) can be implemented using a step-by-step process:

1. Assessment: Identify region-specific ecological degradation and socio-political constraints.
2. Co-Design: Collaborate with local communities and knowledge holders to define goals.
3. Pilot Implementation: Launch small-scale pilots (e.g., PES, agroforestry) based on co-created plans.

4. Monitoring: Use tools like remote sensing or community-based monitoring to assess changes.
5. Scaling: Expand successful pilots and embed practices into governance and policy mechanisms.

For example, in the Eastern Arc Mountains, ANR was scaled through participatory forest monitoring supported by satellite imagery, while in the Andes, landscape mosaics were co-developed via community assemblies and traditional ecological calendars.

4. Challenges and Opportunities

While regenerative forest management offers a compelling vision for ecological renewal and socio-cultural restoration, its implementation in mountain regions is not without significant challenges. However, these obstacles are accompanied by important opportunities for innovation, collaboration, and long-term transformation.

4.1. Challenges

The following challenges were identified:

- Institutional Inertia and Policy Gaps: Many national forestry policies remain oriented toward timber production or static conservation goals. Integrating regenerative objectives requires reconfiguring legal frameworks, incentive structures, and performance indicators.
- Limited Recognition of Indigenous Knowledge: Despite its proven ecological value, Indigenous and local knowledge is often undervalued or excluded from formal decision-making, undermining place-based regeneration potential.
- Fragmented Land Tenure and Governance: In many mountain areas, overlapping land claims, weak enforcement, and ambiguous jurisdiction hinder coherent landscape-scale planning.
- Resource Constraints and Capacity Gaps: Local communities, especially in the Global South, often lack the financial, technical, and institutional support needed to sustain regenerative practices over the long term.
- Climate Uncertainty and Ecological Thresholds: Accelerating climate change introduces high uncertainty. Feedback loops such as drought-fire cycles or glacial retreat may surpass ecological tipping points, complicating regeneration efforts.

4.1.1. Implementation Barriers and Power Asymmetries

Several cases illustrate the challenges of implementing regenerative strategies. In the Khasi Hills, PES schemes initially struggled due to unclear land tenure and unequal benefit sharing. In parts of the Eastern Himalayas, resistance from centralized forestry departments hampered community-based restoration. Power imbalances between international donors and Indigenous stewards can undermine co-creation efforts when funding conditions conflict with traditional knowledge systems.

4.1.2. Navigating Trade-Offs

Trade-offs between ecological and economic objectives are common. For instance, rewilding efforts in the Alps have occasionally conflicted with agricultural or tourism-based livelihoods. In such cases, participatory scenario modeling, multi-stakeholder platforms, and co-benefit planning have helped reconcile competing goals. Recognizing and negotiating these trade-offs is essential for equitable and lasting regeneration.

4.1.3. Emerging Technologies in Regeneration

Technological tools are increasingly important for regenerative forest management. Remote sensing and AI-assisted monitoring can track biodiversity, canopy cover, and landscape connectivity. Blockchain has enabled transparent PES schemes and equitable benefit-sharing. Integrating these tools with community knowledge enhances efficiency while supporting inclusive governance.

4.2. Opportunities

Policy Windows and Global Agendas: The UN Decade on Ecosystem Restoration, post-2020 Global Biodiversity Framework, and growing interest in nature-based solutions provide platforms for mainstreaming regenerative approaches.

Revitalization of Biocultural Knowledge: There is renewed interest in Indigenous land management, spiritual ecologies, and customary governance systems. Supporting these systems enhances both ecological resilience and cultural continuity.

- Innovation in Governance Models: Polycentric and adaptive governance frameworks offer scalable pathways for landscape-level cooperation, particularly in transboundary mountain regions.
- Green Finance and PES Mechanisms: Emerging funding instruments such as green bonds, carbon markets, and payment for ecosystem services can mobilize resources for long-term regeneration.
- Youth and Intergenerational Leadership: Regenerative movements are increasingly driven by younger generations who bring new values, technologies, and cross-cultural solidarity to ecological restoration.

Framing these challenges and opportunities together highlights the urgency, and potential, for mountain forestscapes to serve as testing grounds and exemplars of regenerative transformation in the Anthropocene.

5. Discussion

The analysis presented in this paper contributes to a growing body of scholarship that calls for a shift from conventional sustainability models toward regenerative paradigms, particularly in ecologically and culturally sensitive mountain forestscapes. By integrating diverse knowledge systems—including scientific literature, Indigenous practices, and applied case studies—this work affirms the need for a relational and systems-based approach to ecological renewal.

The results support the argument that sustainability, while foundational, often operates within frameworks of harm reduction, static baselines, and limited adaptability. In contrast, regenerative approaches embrace dynamism, complexity, and co-evolution, offering greater potential to respond to the compound effects of climate change, biodiversity loss, and socio-political marginalization. These findings align with recent regenerative ecology frameworks, including those proposed by Mang and Reed [52], Wahl [45], and Capra and Luisi [70], all of whom emphasize the reweaving of ecological and cultural relationships as central to systemic renewal.

Moreover, the study highlights that regenerative outcomes are deeply place-based, shaped by local governance systems, ecological histories, and cultural meanings. This reinforces calls within the resilience literature to embed restoration within context-specific socio-ecological feedback loops and adaptive governance structures [57,71,72]. Evidence from the Alps, Andes, Eastern Arc, and Carpathians illustrates how decentralized management, biocultural knowledge, and participatory design can deliver resilient landscapes that serve both ecological and human well-being.

Yet, significant barriers remain. Institutional rigidity, fragmented governance, and epistemic injustice continue to constrain regeneration efforts. These challenges suggest a need for institutional transformation that includes legal pluralism, capacity building in community stewardship, and the formal recognition of Indigenous land rights [11,28,71,72]. Furthermore, regeneration must be understood not as a singular goal but as a continuous, evolving process, one that embraces uncertainty, fosters learning, and nurtures relationships between people and ecosystems.

Looking ahead, three cross-cutting priorities emerge:

Integrating science and Indigenous knowledge through co-produced frameworks that validate multiple epistemologies and emphasize reciprocity.

Operationalizing regenerative metrics that capture ecological function, cultural vitality, and governance responsiveness.

Scaling regenerative governance via nested, polycentric institutions that enable landscape-level coordination without undermining local autonomy.

In sum, this paper contributes to an emerging regenerative turn in forest and landscape management, offering theoretical clarity, practical pathways, and critical reflections on the systems transformations needed to sustain mountain forestscapes into the future.

While conventional SFM has evolved to include adaptive and participatory elements, its application often remains constrained by technocratic and sectoral silos. Our findings suggest that regeneration, by integrating biocultural systems and adaptive governance, offers a deeper response to complex forest degradation.

6. Conclusions and Recommendations

This paper has examined the conceptual evolution and applied relevance of regenerative approaches to mountain forestscape management, drawing on a synthesis of scientific literature, Indigenous knowledge, and regionally grounded case studies. The findings underscore a growing recognition that traditional sustainability paradigms, while critical for limiting degradation, are insufficient to reverse deep ecological damage or foster long-term socio-ecological resilience in mountain regions.

Key priorities for future action include:

- Integrating science and Indigenous knowledge through co-produced frameworks that validate multiple epistemologies and emphasize reciprocity;
- operationalizing regenerative metrics that capture ecological function, cultural vitality, and governance responsiveness;
- scaling regenerative governance via nested, polycentric institutions that enable landscape-level coordination without undermining local autonomy.

Instead, regenerative strategies emphasize the dynamic, relational, and co-evolutionary character of mountain landscapes. Through practices such as ecological connectivity restoration, assisted natural regeneration, community-led watershed management, and cultural landscape stewardship, regeneration offers a transformative framework for restoring vitality, complexity, and resilience. It is not merely about recovery but about enabling landscapes and communities to thrive in the face of ongoing change.

The challenges to regeneration, ranging from governance fragmentation and policy inertia to resource gaps and climate instability, are substantial. Yet, there are equally significant opportunities, including alignment with global policy windows, innovative finance, and revitalized biocultural knowledge systems. These offer entry points for reshaping institutions and re-centering Indigenous and local leadership in regeneration.

Within the next five years, national and regional forestry frameworks should formally adopt Indigenous co-management protocols, ensuring recognition of customary tenure and land-based knowledge. By 2027, at least 10 community-led forest restoration pilots

should be launched across Natura 2000 mountain areas. Landscape plans in the Alps and Carpathians must be updated by 2026 to embed regenerative practices—including mosaic agroforestry and rewilding corridors. A biocultural monitoring framework, co-developed with local stewards, should be implemented in five pilot sites by 2025. To support this transition, an EU Regenerative Landscapes Fund should be established by 2026, modeled after the existing EU Biodiversity Strategy and LIFE Programme. This fund could support landscape-scale pilot projects and certification of regenerative forest managers. Similar funds should be developed for non-EU regions such as the Andes, Himalayas, and Southeast Asia, potentially through mechanisms like the Global Environment Facility or regional green finance initiatives. Finally, 1000 forest managers should be certified in regenerative and climate-resilient forestry techniques by 2025. These targets are practical, time-bound, and aligned with the EU Green Deal and IPBES Nature Futures Framework.

Among specific targets belong:

- Embed regeneration into national and regional forest policies, moving beyond sustainability metrics to support long-term ecosystem healing and socio-cultural resilience.
- Support and scale Indigenous-led and community-based forest governance, recognizing customary tenure, traditional ecological knowledge, and cultural relationships with land.
- Invest in landscape-scale ecological restoration, prioritizing connectivity corridors, biodiversity refugia, and watershed integrity through place-based planning.
- Develop regenerative finance mechanisms, including blended finance models, carbon markets with social safeguards, and PES schemes that are community controlled and equity oriented.
- Enhance cross-scalar governance and adaptive institutions, fostering partnerships across public, private, and civil society sectors, and ensuring responsiveness to feedback and local knowledge.
- Promote interdisciplinary and transdisciplinary research, bridging ecological science, Indigenous epistemologies, and policy design to support learning-based regeneration.

By embracing regeneration not as a fixed outcome but as an evolving process rooted in co-evolutionary relationships, mountain forestscapes can become exemplary sites of resilience building. Over the next five years, strategic actions, such as launching community-led pilots, embedding co-management into forestry frameworks, and establishing regenerative finance mechanisms, can catalyze this shift. These priorities align with the IPBES Nature Futures Framework and EU Green Deal while remaining flexible enough to adapt to diverse global mountain contexts.

Funding: This research was funded by the Integrated Technological and Information Platform for Wildfire Management (SILVANUS) H2020 Green Deal, project No. 101037247.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ANR	Assisted natural regeneration
CFUG	Community Forest User Group
IPBES	Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services
NbS	Nature-based solutions
NGO	Non-governmental organization
PES	Payment for ecosystem services
SFM	Sustainable forest management
UN	United Nations

Appendix A

Table A1 provides an overview of case studies and ILK sources used in the analysis, including selection rationale and key attributes.

Table A1. Case and ILK source summary.

Case Study Region	Intervention Type	ILK Source Type	Selection Criteria	Key Attributes
Andes	Mosaic agroforestry	Community-authored reports, peer-reviewed ethnographies	Peer-reviewed, cited in FAO and IPBES	Emphasis on food sovereignty, Andean ecological calendars
Alps	Rewilding corridors	NGO reports, co-management studies	Used by EU-funded Rewilding Europe	Restoration of predator corridors, stakeholder integration
Eastern Arc	ANR and forest bylaws	Participatory mapping, local NGO co-authored reports	Recognized by national forest programs	Customary fire and forest use norms
Khasi Hills	Sacred groves and PES	Clan-based governance literature, REDD+ pilot evaluations	Co-developed with Indigenous organizations	Integration of carbon financing and sacred landscape rituals

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